

$$(x+2)(x^2+2x+2)=0$$

Exercise:

Find the particular integral
of each of the following.

(a) $y'''(t) + 2y''(t) + y'(t) + 2y = 8$

Solution:

$$a_n = 2, b = 8$$

The particular integral of this equation:

$$y_p = \frac{b}{a_n}$$

$$\because a_n \neq 0$$

$$y_p = \frac{8}{2} = \boxed{4}$$

$$\underline{\underline{(b)}} \quad y'''(t) + y''(t) + 3y'(t) = 1$$

Solution:

$$a_{n-1} = 3, b = 1$$

The particular integral of this equation:

$$y_p = \frac{b}{a_{n-1}} t$$

$$\because a_n = 0, a_{n-1} \neq 0$$

$$\boxed{y_p = \frac{1}{3} t}$$

$$\underline{\underline{(c)}} \quad 3y'''(t) + 9y''(t) = 1$$

Solution:

$$a_n = 0, a_{n-1} = 0$$

The particular integral of this equations:

$$y_p = kt^2$$

$$y'(t) = 2kt$$

$$y''(t) = 2k$$

$$y'''(t) = 0$$

Hence,

$$(d) y^{(4)}(t) + y''(t) = 4$$

$$3y'''(t) + 9y''(t) = 1$$

$$3(0) + 9(2k) = 1$$

$$18k = 1$$

$$k = \frac{1}{18}$$

Putting the value in equation (i)

$$y_p = kt^2$$

$$y_p = \frac{1}{18}t^2 \quad \underline{\text{Ans}}$$

Solution:

$$y_p = kt^2$$

$$\because a_n = 0, a_{n-1} = 0$$

So,

$$y_p = kt^2 \quad \text{--- equation (1)}$$

$$y'(t) = 2kt$$

$$y_p = 2kt$$

$$y''(t) = 2k$$

$$y'''(t) = 0$$

$$y^{(4)}(t) = 1$$

Hence,

$$4 = 2K$$

$$\frac{2K}{2} = K$$

$$\boxed{K = 2}$$

Putting the value of K in the equation (1)

$$y_p = Kt^2$$

$$\boxed{y_p = 2t^2}$$

Q.2

Find the y_p and the y_c (and hence the general solution) of:

$$(a) y''(t) - 2y'(t) - y(t) + 2y = 4$$

$$[\text{Hint: } x^3 - 2x^2 - x + 2 = (x-1)(x+1)(x-2)]$$

Solution:

$$a_n = 2, b = 4$$

The particular integral of this equation

$$y_p = \frac{b}{a_n}$$

$$y_p = \frac{4x^2}{2} = \underline{2}$$

Its characteristic form is

$$x^3 - 2x^2 - x + 2 = 0$$

Solve by synthetic division

1	1	2	-1	2
	↓	1	-1	-2
	1	-1	-2	0

Factorization Formula:

$$x^2 - x - 2 = 0$$

$$x^2 - 2x + x - 2 = 0$$

$$x(x - 2) + 1(x - 2) = 0$$

$$(x + 1)(x - 2) = 0$$

$$\lambda + 1 = 0, \lambda - 2 = 0$$

$$\boxed{\lambda = -1} \rightarrow \boxed{\lambda = 2}$$

$$Y_c = ?$$

Roots are real and distinct:

$$Y_c = A_1 e^{1t} + A_2 e^{2t}$$

$$Y_c = A_1 e^{-1t} + A_2 e^{2t}$$

General Solution:

$$Y_t = Y_c + Y_p$$

$$Y_t = A_1 e^{-1t} + A_2 e^{2t} + 2$$

$$(b) Y'''(t) + 7Y''(t) + 15Y'(t) + 9Y = 0$$

Solution

$$Y_p = \frac{b}{an}$$

$$an \neq 0$$

$$Y_p = \frac{0}{9} \Rightarrow \boxed{Y_p = 0}$$

Its characteristics equation is,

$$s^3 + 7s^2 + 15s + 9 = 0$$

Solve by Synthetic Division:

-1	1	7	15	9
	↓	-1	-6	-9
	1	6	9	0

$$s^2 + 6s + 9 = 0$$

Solve this by Quadratic formula:

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Hence,

$$a = 1, b = 6, c = 9$$

$$s = \frac{-6 \pm \sqrt{(6)^2 - 4(1)(9)}}{2(1)}$$

$$\lambda = \frac{-6 \pm \sqrt{36-36}}{2}$$

$$\lambda = \frac{-6 \pm \sqrt{0}}{2}$$

$$\lambda_1 = \frac{-6 + \sqrt{0}}{2}, \quad \lambda_2 = \frac{-6 - \sqrt{0}}{2}$$

$$\lambda_1 = \frac{-6 + 0}{2}, \quad \lambda_2 = \frac{-6 - 0}{2}$$

$$\lambda_1 = \frac{-6^3}{2^1}, \quad \lambda_2 = \frac{-6^3}{2^1}$$

$$\boxed{\lambda_1 = -3}$$

$$\boxed{\lambda_2 = -3}$$

$$y_c = ?$$

Roots are repeated case:

$$y_c = A_1 t e^{\lambda_1 t} + A_2 t^2 e^{\lambda_2 t}$$

General Solution:

$$y(t) = y_c + y_p$$

$$(c) \ y'''(t) + 6y''(t) + 10y'(t) + 8y = 8$$

$$[\text{Hint: } \lambda^3 + 6\lambda^2 + 10\lambda + 8 = (\lambda + 4)(\lambda^2 + 2\lambda + 2)]$$

Solution:

$$a_n = 8, \ b = 8$$

The particular integral of this equation

$$y_p = \frac{b}{a_n}$$

$$y_p = \frac{8}{8} = \boxed{1}$$

Its characteristic equation is:

$$\lambda^3 + 6\lambda^2 + 10\lambda + 8 = 0$$

Solve by synthetic Division:

-4	1	6	10	8
↓	-4	-8	-9	
1	2	2	1	0

$$x^2 + 2x + 2 = 0$$

Solve by Quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Hence,

$$a = 1, b = 2, c = 2$$

$$x = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(2)}}{2(1)}$$

$$x = \frac{-2 \pm \sqrt{4 - 8}}{2}$$

$$x = \frac{-2 \pm \sqrt{-4}}{2}$$

$$x = \frac{-2 \pm 2\sqrt{-1}}{2}$$

$$x = \frac{-2 \pm 2i}{2}$$

$$\therefore i = \sqrt{-1}$$

$$x = \underline{\underline{\cancel{2}(-1 \pm i)}} \quad \cancel{2}$$

$$x = -1 \pm i$$

$$s^1 = -1 + i, \quad s^2 = -1 - i$$

$$Y_c = ?$$

Roots are complex case:

$$Y_c = e^{ht} (A_1 \cos vt + A_2 \sin vt)$$

General Solution:

$$Y_t = Y_c + Y_p$$



Convergence and the Routh Theorem:

The solution of a high-degree characteristics equation is not always an easy task. For this reason, it should be of tremendous

help if we can find a way of ascertaining the convergence or divergence of a time path without having to solve for the characteristics roots.

This method is to be found in the Routh theorem, which states that:

The real parts of all of the roots of the n th-degree polynomial equation

$$a_0 s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n = 0$$

are negative if and only if the first n of the following sequence of determinants:

a_1	a_1	a_3	a_1	a_2	a_5	a_1	a_3	a_5	a_7
a_0	a_0	a_2	a_0	a_1	a_4	a_0	a_2	a_4	a_6
	a_0	a_2	0	a_0	a_3	0	a_1	a_3	a_5
						0	a_0	a_2	a_4

all are positive.

In applying this theorem, it should be remembered that $|a_{ii}| = 0$.

Further, it is to be understood that we should take $a_{im} = 0$ for all $m > n$.

Determinants depend upon the number of n^{th} e.g. if $n=3$ then we take 3 determinant.

If the all values of the determinant result will be positive then we can say it is the case of convergence.

Example:

Fourth Theorem

Solution:-

$$n=4$$

Coefficients are,

$$a_0 = 1, a_1 = 6, a_2 = 14, a_3 = 16, a_4 = 8$$

$$\text{and } a_5 = a_6 = a_7 = 0$$

First four Determinants:

$$|a_1| = \boxed{6}$$

$$\begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix} = \begin{vmatrix} 6 & 16 \\ 1 & 14 \end{vmatrix} \\ = (6 \times 14) - (1 \times 16) \\ = 84 - 16 \\ = \boxed{68}$$

$$\begin{vmatrix} a_1 & a_3 & a_5 \\ a_0 & a_2 & a_4 \\ 0 & a_1 & a_3 \end{vmatrix} = \begin{vmatrix} 6 & 16 & 0 \\ 1 & 14 & 8 \\ 0 & 6 & 16 \end{vmatrix}$$

$$= 6 \begin{vmatrix} 14 & 8 \\ 6 & 16 \end{vmatrix} - 16 \begin{vmatrix} 1 & 8 \\ 0 & 16 \end{vmatrix} + 0$$

$$= 6(14 \times 16 - 6 \times 8) - 16(1 \times 16 - 0 \times 8) + 0$$

$$= 6(224 - 48) - 16(16) + 0$$

$$= 6(176) - 256$$

$$= 1056 - 256$$

$$= \boxed{800}$$

$$\begin{array}{c|cccc}
 a_1 & a_3 & a_5 & a_7 \\
 a_0 & a_2 & a_4 & a_6 \\
 0 & a_1 & a_3 & a_5 \\
 0 & a_0 & a_2 & a_4
 \end{array}
 \begin{array}{c}
 6 \quad 16 \quad 0 \quad 0 \\
 1 \quad 14 \quad 8 \quad 0 \\
 0 \quad 6 \quad 16 \quad 0 \\
 0 \quad 1 \quad 14 \quad 8
 \end{array}$$

$$= 6 \begin{array}{c|ccc}
 14 & 8 & 0 \\
 6 & 16 & 0 \\
 1 & 14 & 8
 \end{array}
 \begin{array}{c}
 -16 \\
 0 \\
 0
 \end{array}
 \begin{array}{c|ccc}
 1 & 8 & 0 \\
 0 & 16 & 0 \\
 0 & 14 & 8
 \end{array}
 \begin{array}{c}
 +0 \\
 +0 \\
 +0
 \end{array}$$

$$= 6 \left[\begin{array}{c|c|c|c|c|c|c|c|c|c|c|c|c|c}
 14 & 16 & 0 & -8 & 6 & 0 & +0 & -16 & 1 & 16 & 0 & -8 & 0 & 0 & +0 \\
 14 & 8 & & & 1 & 8 & & & 14 & 8 & & 0 & 8 & &
 \end{array} \right]$$

$$= 6 [14(16 \times 8 - 14 \times 0) - 8(6 \times 8 - 1 \times 0)] - 16 [1(16 \times 8 - 14 \times 0) - 8(0 \times 8 - 0 \times 0)]$$

$$= 6 [14(128) - 8(48)] - 16 [1(128) - 8(0)]$$

$$= 6(1792 - 384) - 16(128 - 0)$$

$$= 6(1408) - 16(128)$$

$$= 8448 - 2048$$

$$= \boxed{6,400}$$

They are all the values is positive and the time path is convergent.

Exercise

Without finding their characteristics roots, determine whether the following differential equations will give rise the convergent time path.

$$(a) y'''(t) - 10y''(t) + 27y'(t) - 18y = 3$$

Solution:

$$a_0 = 1, a_1 = -10, a_2 = 27, a_3 = -18$$

and $a_4 = a_5 = a_6 = 0$

$$|a_1| = \boxed{-10}$$

The time path is not convergent.

$$(b) y'''(t) + 11y''(t) + 34y'(t) + 24y = 5$$

Solution:

$$a_0 = 1, a_1 = 11, a_2 = 34, a_3 = 24,$$

and $a_4 = a_5 = a_6 = 0$

$$|a_1| = \boxed{11}$$

$$\begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix} = \begin{vmatrix} 11 & 24 \\ 1 & 34 \end{vmatrix} = (11 \times 34) - (1 \times 24)$$

$$= (374) - 24$$

$$= \boxed{350}$$

$$\begin{vmatrix} a_1 & a_3 & a_5 \\ a_0 & a_2 & a_4 \\ 0 & a_1 & a_3 \end{vmatrix} = \begin{vmatrix} 11 & 24 & 0 \\ 1 & 34 & 0 \\ 0 & 11 & 24 \end{vmatrix}$$

$$= 11 \begin{vmatrix} 34 & 0 & -24 \\ 11 & 24 & 0 \end{vmatrix} + 0 \begin{vmatrix} 1 & 0 \\ 0 & 24 \end{vmatrix} + 0$$

$$= 11(34 \times 24) - (11 \times 0) - 24(1 \times 24 - 0 \times 0) + 0$$

$$= 11(816) - 24(24)$$

$$= 8976 - 576$$

$$= \boxed{8400}$$

The time path is convergent.

$$(c) y'''(t) + 4y''(t) + 5y'(t) - 2y = -2$$

Solution:

$$a_0 = 1, a_1 = 4, a_2 = 5, a_3 = -2,$$

$$\text{and } a_4 = a_5 = a_6 = 0$$

$$|a_1| = \boxed{4}$$

$$\begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix} = \begin{vmatrix} 4 & -2 \\ 1 & 5 \end{vmatrix}$$

$$= 4 \times 5 - 1 \times (-2)$$

$$= \boxed{22}$$

$$\begin{vmatrix} a_1 & a_3 & a_5 \\ a_0 & a_2 & a_4 \\ 0 & a_1 & a_3 \end{vmatrix} = \begin{vmatrix} 4 & -2 & 0 \\ 1 & 5 & 0 \\ 0 & 4 & -2 \end{vmatrix}$$

$$= 4 \begin{vmatrix} 5 & 0 \\ 4 & -2 \end{vmatrix} - (-2) \begin{vmatrix} 1 & 0 \\ 0 & -2 \end{vmatrix} + 0$$

$$= 4(5 \times (-2) - 4 \times 0) + 2(1 \times (-2) + 0 \times 0) + 0$$

$$= 4(-10) + 2(-2)$$

$$= -40 - 4$$

$$= \boxed{-44} \rightarrow \text{Disconvergent}$$